

Validation Cross-Criterion Integration Module (VCCIM)

Architecture Ecosystem: Structured Reality Standards™
Architecture Family: VALIDOS® — Validation Governance Architecture
Parent Standard: Validation Determination Standard (VDTS)
Operational Layer: Validation Determination Governance Layer
Category: AI & Interpretation
Subcategory: Validation Governance
Type: Parent Standard Module
Governed Space: Validation Cross-Criterion Integration
Version: 1.0
Status: Canonical · Open Module
Effective Date: 11 August 2026
Compatibility: OOF® Methodology OS™ · GOA™ · OBIDENITY® · INTEGROS®
· ORA™ · AGA™ · AIG® · CLIA® · MGIA™ · ASGA™ · RIS™
AI-Readable: Yes
Authority: OOF®
Protection: MIP® — Methodological Intellectual Property
Canonical Language: English (UCL™)

Governed Space

Validation Cross-Criterion Integration

Canonical Definition

Validation Cross-Criterion Integration Module (VCCIM) defines the governance framework for combining, relating, ordering, constraining, and interpreting multiple criterion-level Validation Determinations within a unified determination structure while preserving mandatory criteria, critical criteria, dependencies, exclusions, conditionality, hierarchical relationships, conflict constraints, uncertainty, scope boundaries, and criterion-specific methodological significance.

It establishes the conditions under which individual criterion outcomes may legitimately contribute to an overall Validation Determination without being flattened into an unjustified average, score, vote, or simplified pass/fail result.

Operational Role

VCCIM serves as the fourth internal governance space of the Validation Determination Standard.

The preceding modules establish:

VDBRQM — Which results may legitimately participate in determination?

VCDM — What does the qualified result set demonstrate about each individual Validation Criterion?

VDCUGM — Which conflicts, uncertainties, and Determination Constraints must remain visible?

VCCIM asks:

How should the resulting criterion determinations be structurally combined so that their distinct governance significance is preserved in the overall validation conclusion?

The governed progression is:

Criterion Determinations → Criterion Relationship Mapping → Mandatory & Critical Logic → Dependency & Exclusion Logic → Conditionality Propagation → Conflict Constraint Integration → Hierarchical Integration → Cross-Criterion Determination Structure

VCCIM does not issue the final overall Validation Determination.

It establishes the governed integration structure from which that final determination can be legitimately derived.

Module Operational Space

VCCIM governs:

- cross-criterion integration,
- criterion relationship mapping,
- criterion hierarchy,
- criterion grouping,
- mandatory criteria,
- critical criteria,
- supporting criteria,
- dependent criteria,
- prerequisite criteria,
- exclusionary criteria,
- conditional criteria,
- independent criteria,
- criterion interaction,
- criterion weighting,
- weighting governance,
- non-compensatory criteria,
- compensatory criteria,
- criterion aggregation,
- integration logic,
- integration rules,
- cross-criterion conflict,
- Determination Constraint integration,
- uncertainty propagation,

- conditionality propagation,
- scope propagation,
- partial integration,
- hierarchical integration,
- integration consistency,
- integration completeness,
- integration change,
- and integration traceability.

Module Function

VCCIM applies after individual criteria have received governed determinations and material conflicts and uncertainties have been classified.

Its function is to prevent:

- a large number of satisfied minor criteria masking failure of one critical criterion,
- mandatory criteria being diluted by favorable supporting criteria,
- criteria with different scopes being combined as though they applied universally,
- conditional satisfaction losing its conditions,
- Not Determinable criteria being silently treated as satisfied,
- criterion dependencies being ignored,
- prerequisite failure being overlooked,
- exclusion conditions being averaged away,
- post-hoc weighting being used to obtain a preferred result,
- incompatible criterion outcomes being combined without governance,
- and overall validation being reduced to an arbitrary percentage.

Criterion Integration Is Not Simple Aggregation

VALIDOS® establishes:

Criterion Integration ≠ Criterion Counting

and:

Criterion Integration ≠ Arithmetic Averaging

A validation containing:

9 Satisfied Criteria + 1 Critical Not Satisfied Criterion

does not necessarily equal:

90% Validated

Such a representation may destroy the actual governance meaning of the validation.

VCCIM therefore integrates criteria according to their defined

relationships and significance.

Criterion Relationship Map

Before integration, VCCIM establishes how criteria relate to one another.

A Criterion Relationship Map may distinguish:

- independent criteria,
- dependent criteria,
- mandatory criteria,
- critical criteria,
- supporting criteria,
- prerequisite criteria,
- exclusionary criteria,
- conditional criteria,
- hierarchical criteria,
- and grouped criteria.

These relationships should derive from the governed Validation Criteria Framework rather than being invented after results are known.

Independent Criteria

Independent criteria may be determined separately and may contribute independently to the overall Validation Determination.

Independence does not imply equal significance.

Dependent Criteria

A Dependent Criterion relies upon one or more other criteria.

For example:

Criterion B is meaningful only if Criterion A is Satisfied.

Where Criterion A is Not Satisfied or Not Determinable, VCCIM must govern the effect on Criterion B.

Prerequisite Criteria

A Prerequisite Criterion must be established before another criterion or criterion group can legitimately support an overall conclusion.

Failure of a prerequisite may make downstream positive findings unusable for integration.

Mandatory Criteria

Mandatory criteria establish non-optional validation conditions.

Where a mandatory criterion is Not Satisfied, the permitted overall determination may be constrained according to the approved integration logic.

Where a mandatory criterion is Not Determinable, the architecture may likewise prevent unrestricted positive determination where the criteria framework requires positive establishment.

Critical Criteria

Critical criteria possess disproportionate consequence.

Their outcomes may trigger:

- overall negative determination,
- conditional determination,
- additional validation,
- restricted scope,
- or overall indeterminacy

depending upon the governed criteria logic.

VCCIM preserves criticality rather than allowing it to disappear within aggregation.

Supporting Criteria

Supporting criteria add evidentiary or interpretive strength but may not independently control the overall determination.

They may:

- reinforce confidence,
 - provide context,
 - reduce uncertainty,
 - support one determination pathway,
 - or contribute to hierarchical criterion groups.
-

Exclusionary Criteria

An Exclusionary Criterion establishes a condition that prevents a specified overall conclusion regardless of favorable results elsewhere.

For example:

Presence of prohibited catastrophic behavior → Positive Overall Determination Not Permitted

VCCIM preserves such exclusion logic.

Conditional Criteria

A Conditional Criterion becomes relevant only when defined conditions exist.

For example:

If autonomous authority exceeds Level X → Criterion Y becomes mandatory.

Conditional applicability must be resolved before integration.

Criterion Hierarchy

Complex validation structures may contain multiple levels.

For example:

Metric → Criterion → Criterion Group → Validation Domain → Overall Validation

VCCIM governs upward integration across these levels.

A higher-level determination should remain traceable to its lower-level criterion basis.

Criterion Grouping

Criteria may be grouped according to:

- capability,
- safety domain,
- system component,
- governance function,
- operational environment,
- population,
- lifecycle phase,
- or another governed structure.

Group-level determinations may support system-level interpretation without collapsing individual criterion identities.

Cross-Criterion Dependency

Criterion outcomes may interact.

Examples include:

- one criterion activates another,
- one criterion invalidates another's positive contribution,
- two criteria must both be satisfied,
- at least one of several alternative criteria must be satisfied,
- or one criterion changes the permitted interpretation of another.

VCCIM governs these relationships explicitly.

AND Logic

Some criteria structures require:

Criterion A AND Criterion B AND Criterion C

for a positive integration outcome.

Where one mandatory element fails, favorable performance elsewhere does not satisfy the integration rule.

OR Logic

Some structures may permit:

Criterion A OR Criterion B

where either legitimate pathway can satisfy the higher-level requirement.

The permitted alternatives must be defined in advance.

Conditional Logic

Integration may use structures such as:

IF A → B Required

IF C Not Satisfied → D Cannot Be Positive

or:

IF Environment X → Criterion E Applies

Conditional relationships must remain explicit.

Non-Compensatory Criteria

A **Non-Compensatory Criterion** cannot have its failure offset by favorable results from other criteria.

Critical safety requirements are common examples.

VCCIM establishes:

Strong Performance Elsewhere Cannot Compensate for a Non-Compensatory Failure

where the criteria framework defines such logic.

Compensatory Criteria

Some criteria may legitimately permit trade-offs.

For example, an overall performance domain may combine several metrics where stronger performance in one dimension can offset weaker performance in another within governed boundaries.

Where compensatory logic exists, it must be explicit and methodologically justified.

Weighting Governance

Where criterion weighting is used, VCCIM governs:

- weight identity,
- weight rationale,
- weight version,
- criterion association,
- permitted aggregation model,
- sensitivity,
- and change control.

Weights should normally be established before outcome interpretation.

Post-Hoc Weight Manipulation

VALIDOS® rejects:

Observed Results → Change Weights → Desired Outcome

Weight modification after results are known may be legitimate only where a methodological error is identified and the change is governed through reassessment.

Historical weighting must remain visible.

Weighted Integration

Where weights are legitimate, VCCIM may permit weighted integration.

However, weighting must not override:

- mandatory criteria,
- critical criteria,
- exclusionary conditions,
- Determination Constraints,
- or non-compensatory rules

unless the criteria framework explicitly permits it.

Score-Based Integration

Some validation systems may use scores.

VCCIM permits scoring where the score has a governed methodological meaning.

It rejects score use where:

- criterion significance differs materially,
 - mandatory conditions are hidden,
 - uncertainty is omitted,
 - failure thresholds are obscured,
 - or scores create false precision.
-

Percent Validated

VALIDOS® treats statements such as:

“The system is 87% validated.”

with caution.

Such a statement is legitimate only where the percentage has a formally defined and methodologically justified interpretation.

Simple division:

Satisfied Criteria / Total Criteria

does not automatically establish meaningful validation percentage.

Conditionality Propagation

A Conditionally Satisfied criterion must propagate its material conditions upward where the overall conclusion depends upon it.

For example:

Criterion Satisfied only under Environment X

may mean:

Overall Determination positive only under Environment X

if that criterion is necessary for the positive determination.

Condition Interaction

Multiple conditional criteria may create combined conditions.

For example:

Criterion A requires Environment X

and:

Criterion B requires Configuration Y

The resulting integrated determination may require:

Environment X + Configuration Y

where both criteria are necessary.

Condition Conflict

Conditions themselves may conflict.

For example:

Criterion A may be satisfied only under Configuration X.

Criterion B may be satisfied only under Configuration Y.

If X and Y cannot coexist, the object may not support a positive integrated determination even though each criterion can independently be satisfied.

VCCIM makes this visible.

Not Determinable Propagation

A Not Determinable criterion must not silently become neutral.

Its effect depends upon its role.

If the criterion is:

Supporting

it may have limited impact.

If it is:

Mandatory

or:

Critical

it may prevent unrestricted positive determination.

Negative Criterion Propagation

A Not Satisfied criterion also propagates according to governance significance.

Its effect may be:

- local,
 - group-level,
 - domain-level,
 - scope-limiting,
 - condition-triggering,
 - or overall determination-controlling.
-

Conflict Constraint Integration

VDCUGM may produce Determination Constraints.

VCCIM integrates those constraints into criterion combination.

Examples include:

- Criterion A cannot be integrated with Criterion B,
 - Criterion C remains Not Determinable,
 - scope must be reduced,
 - additional validation required,
 - positive system-level conclusion prohibited,
 - or only partial determination allowed.
-

Uncertainty Propagation

Criterion-level uncertainty may affect higher-level integration.

VCCIM does not simply average uncertainty away.

It asks whether:

- uncertainty is independent,
 - uncertainty is shared,
 - uncertainty is cumulative,
 - uncertainty is concentrated in critical criteria,
 - or uncertainty prevents the higher-level conclusion.
-

Cumulative Uncertainty

Several criterion outcomes may each be acceptable individually while collectively creating excessive uncertainty.

VALIDOS® therefore permits assessment of cumulative determination uncertainty across criterion groups.

Shared Uncertainty

Multiple criteria may depend upon the same uncertain evidence, source, method, or execution condition.

Their uncertainties are therefore not fully independent.

VCCIM prevents repeated use of the same uncertain basis from falsely appearing as diversified support.

Cross-Criterion Conflict

Criterion outcomes may be individually valid yet mutually incompatible at a higher governance level.

For example:

Performance Criterion: Satisfied

Safety Boundary Criterion: Not Satisfied

The conflict may not be a contradiction in facts.

It may reveal a legitimate system trade-off that must constrain overall validation.

Scope Propagation

Each criterion may possess its own legitimate scope.

VCCIM integrates only overlapping or explicitly compatible scopes.

An overall determination cannot claim a scope wider than that supported by required criteria.

Conceptually:

Overall Determination Scope = Intersection of Required Criterion Scopes

where appropriate.

Scope Fragmentation

Different criteria may support different valid scopes.

Where no single common scope exists, VCCIM may require:

Partial Validation Determination

rather than one universal conclusion.

Temporal Integration

Criterion outcomes may apply to different time periods.

A higher-level determination must account for temporal compatibility.

A criterion established six months earlier may not automatically integrate with a newly validated criterion if material system changes occurred between them.

Version Integration

Criterion outcomes derived from different Validation Object versions should not be combined automatically.

VCCIM requires sufficient version continuity or explicit transfer justification.

Validation Depth Integration

Required criteria may have been validated at different depths.

An overall determination should not claim a higher level of assurance than the critical criterion basis supports.

The integration may therefore be bounded by the weakest required Validation Depth where methodologically appropriate.

Hierarchical Integration

VCCIM may create intermediate determinations.

For example:

Criterion Outcomes

↓

Safety Domain Determination

Performance Domain Determination

Reliability Domain Determination

↓

System-Level Integration Structure

This improves interpretability for complex validation systems.

Partial Integration

Where only part of the criterion structure can be legitimately integrated, VCCIM preserves partiality.

Possible outputs may identify:

- integrated validated domain,
 - negative domain,
 - unresolved domain,
 - conditional domain,
 - or excluded domain.
-

Integration Completeness

A Cross-Criterion Integration structure may be:

Integration Complete

Conditionally Complete

Partially Complete

Incomplete

or:

Integration Completeness Undetermined

Completeness asks whether all materially required criterion relationships have been addressed.

Integration Consistency

VCCIM assesses whether the integration structure itself remains internally coherent.

Potential inconsistencies include:

- mandatory criterion omitted,
 - incompatible conditions combined,
 - failed prerequisite ignored,
 - wrong weighting applied,
 - exclusion rule bypassed,
 - criterion version mismatch,
 - or Determination Constraint omitted.
-

Integration Conflict

An Integration Conflict exists where the governed criterion relationship model cannot support one coherent integration outcome without additional governance.

This may require:

- revised scope,
 - partial integration,
 - further validation,
 - criterion reassessment,
 - or overall indeterminacy.
-

Integration Logic Freeze

Before final Overall Validation Determination, the material cross-criterion integration logic should be version-fixed.

This protects against changing:

- weights,
- mandatory status,
- exclusion rules,
- hierarchy,
- dependency logic,
- or integration rules

after seeing the combined outcome.

Integration Logic Change

A material change may be legitimate where:

- criteria architecture error is identified,
 - regulatory requirement changes,
 - methodology is revised,
-

- or criterion relationship was incorrectly specified.

Such change should trigger reassessment rather than silently rewriting the existing determination.

Cross-Criterion Integration Record

A formal record may preserve:

- Criteria Set,
 - Criterion Determinations,
 - criterion roles,
 - mandatory status,
 - critical status,
 - dependencies,
 - prerequisites,
 - exclusionary conditions,
 - grouping,
 - hierarchy,
 - weights where used,
 - conditions,
 - Determination Constraints,
 - uncertainty,
 - scope,
 - integration rules,
 - integration completeness,
 - integration consistency,
 - and resulting Cross-Criterion Determination Structure.
-

Minimum Implementation Framework

1. Establish Criterion Relationship Map

Identify mandatory, critical, supporting, dependent, prerequisite, exclusionary, conditional, hierarchical, and independent criteria.

2. Import Criterion Determinations

Preserve each criterion's:

- status,
- scope,
- depth,
- conditions,
- limitations,
- conflicts,
- and uncertainty.

3. Apply Mandatory & Critical Logic

Determine which criterion outcomes constrain higher-level integration.

4. Apply Dependency & Exclusion Logic

Respect prerequisites, conditional activation, exclusion rules, and cross-criterion relationships.

5. Apply Governed Weighting or Aggregation Where Permitted

Use only pre-established or methodologically justified integration rules.

6. Propagate Conditions & Scope

Preserve the limitations necessary for the integrated conclusion.

7. Integrate Conflict & Uncertainty Constraints

Apply all relevant Determination Constraints from VDCUGM.

8. Build Hierarchical or Partial Integration

Where appropriate, create group, domain, subsystem, or partial determination structures.

9. Assess Integration Completeness & Consistency

Identify omitted relationships, incompatible rules, or unresolved integration conflicts.

10. Freeze Integration Logic

Preserve the exact logic entering Overall Validation Determination.

11. Preserve Lifecycle Traceability

Maintain:

- criterion relationships,
- criterion statuses,
- mandatory and critical designations,
- weights,
- dependencies,
- exclusions,
- conditions,
- scope,
- uncertainty,
- Determination Constraints,
- hierarchy,
- integration rules,
- integration status,
- versions,
- changes,
- and sufficient lifecycle continuity.

Governance Outputs

VCCIM may produce:

- Criterion Relationship Map,
- Cross-Criterion Integration Record,
- Mandatory Criteria Integration Record,
- Critical Criteria Integration Record,
- Supporting Criteria Integration Record,
- Criterion Dependency Map,
- Criterion Prerequisite Map,
- Criterion Exclusion Map,
- Conditional Criterion Map,
- Criterion Hierarchy Map,
- Criterion Group Determination,
- Validation Domain Determination,
- Non-Compensatory Criterion Record,
- Compensatory Criterion Record,
- Criterion Weighting Record,
- Integration Rule Set,
- Conditionality Propagation Record,
- Scope Propagation Record,
- Uncertainty Propagation Record,
- Determination Constraint Integration Record,
- Cross-Criterion Conflict Record,
- Partial Integration Record,
- Hierarchical Integration Record,
- Integration Completeness Assessment,
- Integration Consistency Assessment,
- Integration Conflict Record,
- Integration Logic Freeze Record,
- Integration Logic Change Record,
- and Cross-Criterion Determination Structure.

Use Case 1 — Autonomous AI System

Scenario

An autonomous AI system has ten validation criteria.

Eight are Satisfied.

One supporting criterion is Conditionally Satisfied.

One critical emergency-stop criterion is Not Satisfied.

Application

VCCIM does not calculate:

9/10 = 90% validated

Instead, it applies the critical criterion logic established in the Validation Criteria Framework.

Result

The critical failure prevents the favorable criteria count from producing an unrestricted positive integration outcome.

Use Case 2 — Healthcare AI

Scenario

A healthcare AI system satisfies accuracy, robustness, and usability criteria.

The population-safety criterion is Satisfied only for adults, while pediatric validation remains Not Determinable.

Application

VCCIM propagates the population boundary across the integrated criterion structure.

Result

The system may support a positive integration for the adult population while pediatric validation remains outside the positive determination scope.

Use Case 3 — Industrial Autonomous System

Scenario

An industrial autonomous system satisfies operational-performance criteria in Configuration A and safety-recovery criteria only in Configuration B.

The two configurations cannot operate simultaneously.

Application

VCCIM identifies a Condition Conflict.

Result

The criteria cannot be combined into one unconditional positive system-level determination merely because each was individually satisfied under different conditions.

The integration structure preserves the incompatibility for final determination.

Architectural Position

Validation Cross-Criterion Integration is the fourth internal governance space of the **Validation Determination Standard (VDTs)**.

The progression now becomes:

**Determination Basis & Result Qualification → Criterion Determination
→ Conflict & Uncertainty Governance → Cross-Criterion Integration**

VDBRQM establishes:

Which results may participate?

VCDM establishes:

What does each criterion demonstrate?

VDCUGM establishes:

Which conflicts, uncertainties, and constraints must remain visible?

VCCIM establishes:

How can the governed criterion outcomes be combined without destroying their individual significance, conditions, dependencies, scope, or uncertainty?

The fifth and final VDTs module can then determine:

What overall Validation Determination does the complete governed validation record legitimately support?

Foundational Principle

Validation criteria should be integrated according to their governance significance and relationships, not merely counted, averaged, or weighted until a convenient result appears.

Canonical Closing Statement

Validation Cross-Criterion Integration Module establishes the structural integration layer of Validation Determination within VALIDOS® by governing how individual criterion outcomes interact across mandatory, critical, supporting, dependent, prerequisite, exclusionary, conditional, compensatory, non-compensatory, and hierarchical relationships. By preserving criterion significance, integration logic, weighting integrity, conditionality, scope, uncertainty, Determination Constraints, partiality, hierarchy, completeness, consistency, and traceability, VCCIM ensures that the overall validation conclusion emerges from the governed structure of the criteria themselves rather than from simplistic counting,

averaging, or post-hoc aggregation of validation results.

Parent Resources

[→ VALIDOS® Validation Governance Architecture — Validation Governance Layer](#)

[→ VALIDOS® Architecture Map](#)

[→ VALIDOS® Complete Standards & Modules Index](#)

Internal Module Flow

[→ Previous module: Validation Determination Conflict & Uncertainty Governance Module \(VDCUGM\)](#)

[→ Next module: Overall Validation Determination Module \(OVDM\)](#)

Related Documents

[→ Validation Determination Standard](#)

[→ About Validation Determination Standard](#)

OOF® MIP® Protection Notice

OOF® · Protected under MIP® — Methodological Intellectual Property

Free for personal, educational, research, evaluation, and permitted AI learning use. Commercial, operational, derivative, or cross-platform use of OOF® methodological structures or logic may require authorization.

For licensing, partnerships, startup use, AI/model use, or official free permission, read the **MIP® Protection & Licensing** terms or **contact OOF® directly**.

Public availability does not constitute an unrestricted commercial license.

Active Protection & Compatibility Detection applies.

Canonical Source

<https://originopenfoundation.org/>